

Leo Chen
Eagle Habitat Image Analysis Project
April 2026

Comparing Classical Color Thresholding and Deep Semantic Segmentation for Environmental Composition Analysis in Eagle Habitat Photographs

Abstract

This project compares two computer vision approaches for estimating environmental characteristics in eagle habitat photographs: a **classical color-threshold segmentation method** and a **deep learning semantic segmentation method**. The dataset contains 106 JPG images from multiple years, processed using the bottom 65% of each image as the region of interest to reduce sky-dominance and focus more on habitat features. The analysis focused on two primary environmental metrics: vegetation fraction and water fraction. Across all 106 images, the classical and deep learning methods showed only moderate-to-weak agreement. For vegetation fraction, the Pearson correlation was **0.413** and the Spearman correlation was **0.276**, with a mean absolute error of **0.202**. For water fraction, the Pearson correlation was **0.254** and the Spearman correlation was **0.197**, with a mean absolute error of **0.346**. These results show that the two methods do not simply produce interchangeable measurements. Instead, they respond differently to reflections, winter vegetation, snow/ice, shoreline ambiguity, and sky-heavy compositions. The main conclusion is that deep semantic segmentation and classical thresholding measure environmental image characteristics differently, and that visual audit/error analysis is necessary before using these values for ecological or climate-related claims.

Keywords: computer vision, semantic segmentation, image processing, eagle habitat, environmental monitoring, SegFormer, classical thresholding

1. Introduction

Photographs can contain useful environmental information beyond their main subject. In eagle photography, the surrounding scene may include water, vegetation, snow, ice, shoreline, sky, or built structures. Measuring these characteristics across years could potentially help describe changes in habitat context, photography conditions, or scene composition. However, environmental image analysis is difficult because real-world photos are not controlled: lighting changes, scenes vary, reflections distort water surfaces, and winter vegetation may not appear green even though it is still vegetation.

This project began as an attempt to quantify environmental features in eagle photographs across different years. Over time, the focus became more methodological: instead of treating the output of one model as

automatically correct, this project compares two different measurement approaches and studies how they disagree. The first approach is a classical computer vision method based on color and brightness thresholds. The second is a deep learning semantic segmentation method using a **SegFormer** model trained for scene segmentation.

The main research question is:

How do classical color-threshold segmentation and deep semantic segmentation differ when estimating water and vegetation coverage in heterogeneous eagle habitat photographs?

This framing is important because the dataset is not a perfectly controlled repeat-photography dataset. The images vary by year, scene type, lighting, season, and framing. Therefore, the strongest claim is not that the project proves a specific environmental trend. The stronger and more defensible claim is that the project evaluates how two image-analysis methods behave under realistic, messy photographic conditions.

2. Background

Computer vision can be used to extract measurable information from images, including object presence, scene composition, and environmental context. In this project, computer vision is used to estimate how much of each eagle photograph contains water or vegetation. However, this task is difficult because real-world wildlife photographs are not controlled. Lighting, shadows, reflections, snow, ice, sky, and seasonal changes can all affect how an image-analysis method interprets a scene.

Classical image processing methods rely on manually designed rules such as color thresholds, brightness thresholds, saturation thresholds, and simple texture patterns. These methods are useful because they are fast, interpretable, and easy to adjust. However, they are also sensitive to visual conditions. For example, a classical method may miss leafless winter vegetation because it is brown or gray rather than green. It may also confuse reflective water, shadows, or blue-gray surfaces with actual water.

Deep learning semantic segmentation methods approach the problem differently. Instead of using fixed color rules, semantic segmentation models assign a category label to each pixel based on learned visual patterns. In this project, the deep learning method used a SegFormer-based model trained for broad scene segmentation (Xie et al., 2021; Hugging Face, 2026). This allows the model to recognize objects and surfaces semantically, such as trees or water, even when their colors vary. However, the model is still not perfect because it was not trained specifically on eagle habitat photographs.

Because the model used in this project was fine-tuned on ADE20K scene-parsing categories, its labels are broad scene categories rather than eagle-habitat-specific ecological classes (Zhou et al., 2017). This makes the model useful for general scene interpretation, but it also creates possible mismatch when applying it to wildlife photographs with reflections, frozen water, branches, shoreline edges, and sky-heavy compositions.

This background explains why the project compares the two methods rather than assuming one is automatically correct. Classical thresholding and deep semantic segmentation represent different ways of measuring the same scene. One is transparent but fragile, while the other is more semantically flexible but model-dependent. Comparing them helps show how model choice affects environmental measurements extracted from photographs.

3. Methods

3.1 Dataset

The dataset contains 106 JPG eagle photographs stored in the project folder under photos\converted. The images span multiple years, including 2016, 2018, 2019, 2021, 2022, 2023, 2024, and 2025. Capture years were extracted from image metadata where available. Each image was processed using a **region of interest** equal to the **bottom 65%** of the image. This ROI was selected because full-frame eagle photographs often include large areas of sky, which can distort habitat measurements when the goal is to analyze water, vegetation, snow/ice, shoreline, and ground features. The same ROI setting was applied to both methods so that the comparison measured the same portion of each image.

3.2 Classical Segmentation

The classical method uses rule-based image processing. It classifies pixels using color, saturation, brightness, and threshold logic. For example, water is estimated from color/brightness patterns associated with blue or dark water-like regions, while vegetation is estimated from green or brown vegetation-like thresholds. This method is fast and interpretable because its decisions come from explicit rules. However, it is also fragile. It can fail when vegetation is leafless and gray-brown, when water reflects the sky or trees, when snow/ice changes brightness patterns, or when shadows resemble water.

The classical method produced per-image values such as:

- vegetation_frac
- water_frac

These values represent the estimated fraction of the ROI classified as vegetation or water.

3.3 Deep Learning Segmentation

The deep learning method uses semantic segmentation. Instead of relying only on color thresholds, it labels pixels using learned visual patterns from a pretrained SegFormer model. The model’s segmentation labels were grouped into broader environmental categories, including water and vegetation. This approach can recognize semantic objects such as trees even when they are not bright green, and it can sometimes separate water from other surfaces more effectively than thresholding. However, it can still fail when the scene is outside its training distribution, when reflections confuse semantic boundaries, or when snow/ice and shoreline areas are ambiguous.

The DL method also produced per-image water_frac and vegetation_frac values for the same bottom-65% ROI.

3.4 Comparison Metrics

The two methods were compared using:

- Pearson correlation: measures linear agreement between classical and DL estimates.
- Spearman correlation: measures whether the two methods rank images similarly.
- Mean absolute error (MAE): measures the average absolute difference between the two methods.
- Year-level means: compares how each method summarizes vegetation and water by year.
- Outlier audit: identifies images where the two methods disagree most strongly and explains likely failure modes using visual inspection.

The final outputs included per-image CSV files, year-level summary CSV files, comparison plots, and an audit table of the largest disagreements between methods.

4. Results

4.1 Overall Agreement Between Methods

Metric	n	Pearson	Spearman	MAE	Classical Mean	DL Mean
vegetation_frac	106	0.413	0.276	0.202	0.152	0.234
water_frac	106	0.254	0.197	0.346	0.430	0.249

Table 1. Overall comparison between the classical and DL methods across 106 images.

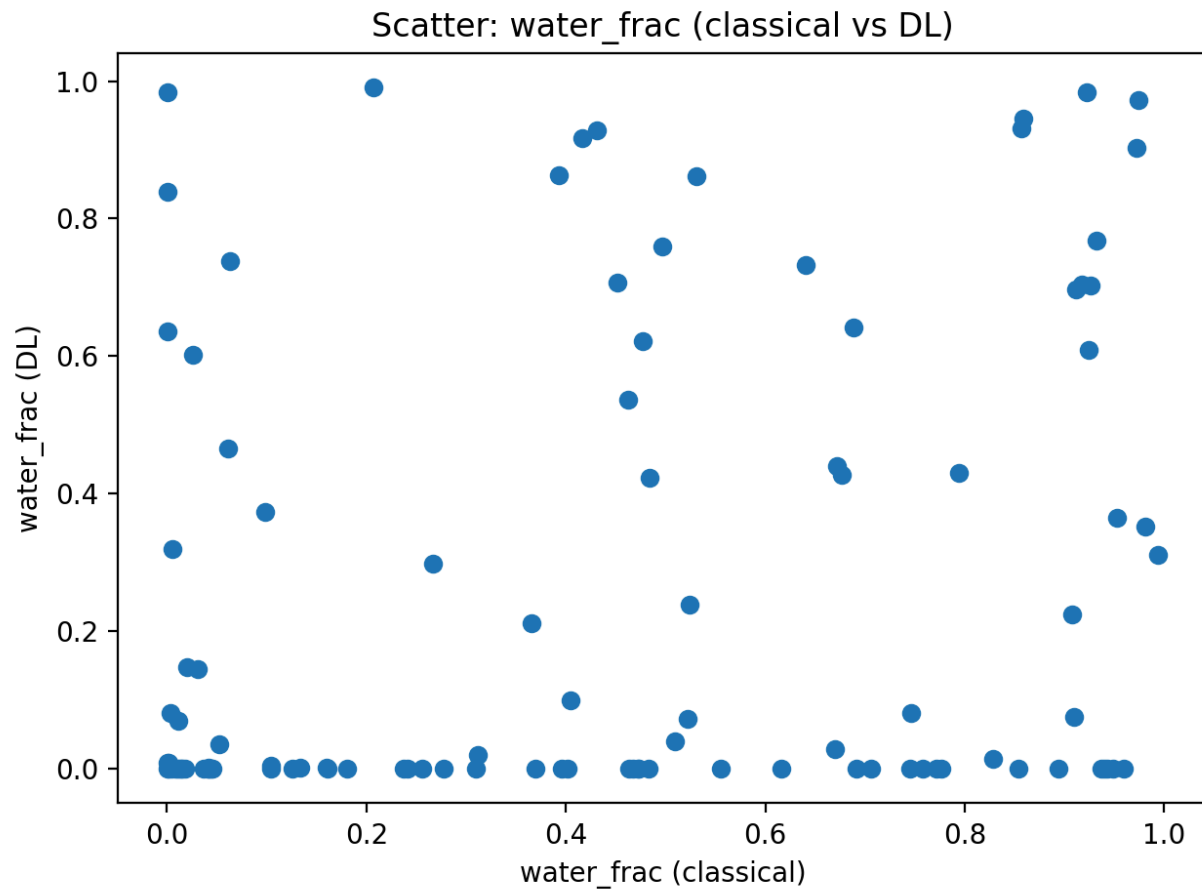


Figure 1. Scatter plot for water fraction. The weak correlation between classical and DL water estimates shows that the two methods often rank images differently in terms of water coverage.

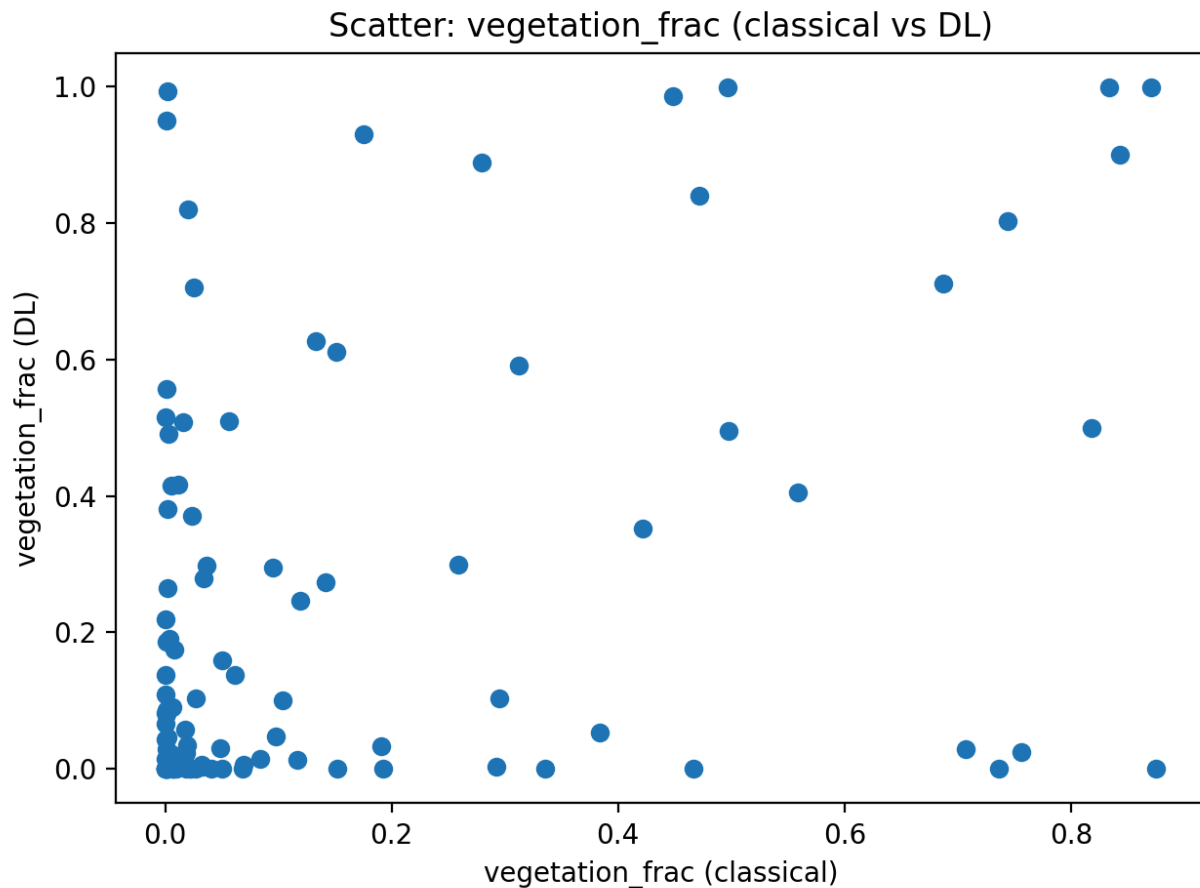


Figure 2. Scatter plot for vegetation fraction. Vegetation agreement was stronger than water agreement but still limited, reflecting the difference between color-based and semantic vegetation detection.

The methods did not strongly agree overall.

For vegetation fraction:

- n = **106 images**
- Pearson correlation = **0.413**
- Spearman correlation = **0.276**
- Mean absolute error = **0.202**
- Classical mean = **0.152**
- DL mean = **0.234**

For water fraction:

- n = 106 images
- Pearson correlation = **0.254**
- Spearman correlation = **0.197**
- Mean absolute error = **0.346**

- Classical mean = **0.430**
- DL mean = **0.249**

These results show that the two approaches are related only weakly to moderately, depending on the metric. Vegetation agreement was stronger than water agreement, but still not high. Water showed the largest disagreement, with the classical method estimating a much higher average water fraction than the DL method. This suggests that the classical method may be more likely to classify reflective, dark, or blue-gray regions as water, while the DL model uses a more semantic interpretation.

The results do not support treating the two methods as interchangeable. Instead, they show that model choice strongly affects the measured environmental composition of the same images.

4.2 Year-Level Vegetation Results

The year-level vegetation estimates differed between methods.

Vegetation fraction by year:

- 2016: Classical = 0.162, DL = 0.159
- 2018: Classical = 0.035, DL = 0.241
- 2019: Classical = 0.080, DL = 0.250
- 2021: Classical = 0.004, DL = 0.225
- 2022: Classical = 0.005, DL = 0.174
- 2023: Classical = 0.187, DL = 0.178
- 2024: Classical = 0.405, DL = 0.431
- 2025: Classical = 0.040, DL = 0.159

The two methods were similar in 2016 and 2023, and both produced high vegetation estimates in 2024. However, they diverged strongly in 2018, 2019, 2021, 2022, and 2025. In those years, the DL method generally estimated more vegetation than the classical method.

A likely explanation is that many eagle habitat photos contain leafless trees, gray-brown branches, and winter vegetation. The classical method depends more heavily on color patterns, so it may miss vegetation that is not green. The DL method can classify tree and branch structures as vegetation even when they are brown or gray. This makes the DL method more semantically flexible, but it also means that it may count winter canopy structure as vegetation even when there is little green leaf cover.

4.3 Year-Level Water Results

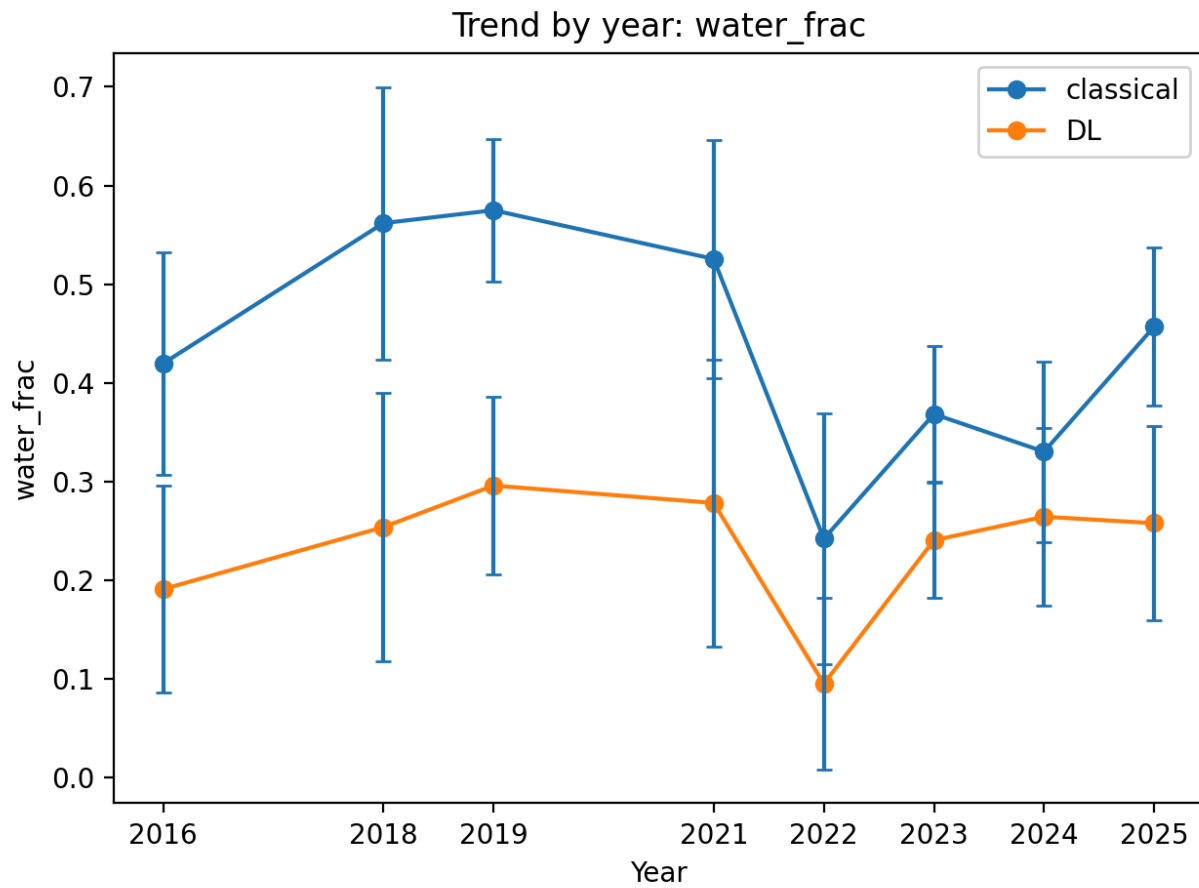


Figure 3. Trend comparison for vegetation fraction. DL often estimated higher vegetation coverage than the classical method, especially in years with leafless or brown vegetation, indicating that semantic segmentation can identify vegetation structure beyond green color.

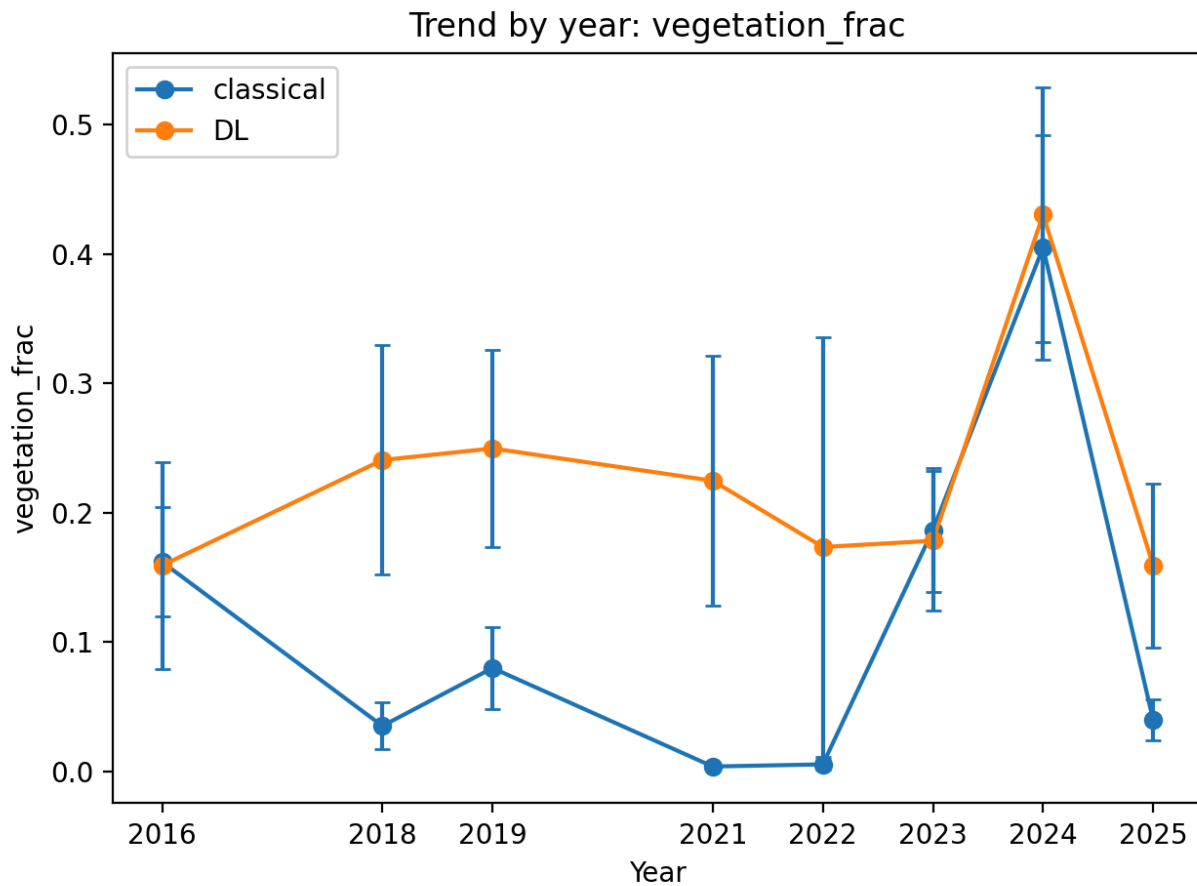


Figure 4. Trend comparison for vegetation fraction. DL often estimated higher vegetation coverage than the classical method, especially in years with leafless or brown vegetation, indicating that semantic segmentation can identify vegetation structure beyond green color.

Water estimates also differed by method.

Water fraction by year:

- 2016: Classical = 0.420, DL = 0.191
- 2018: Classical = 0.562, DL = 0.254
- 2019: Classical = 0.575, DL = 0.296
- 2021: Classical = 0.526, DL = 0.278
- 2022: Classical = 0.242, DL = 0.095
- 2023: Classical = 0.368, DL = 0.241
- 2024: Classical = 0.330, DL = 0.264
- 2025: Classical = 0.457, DL = 0.258

Across every year, the classical method estimated more water than the DL method. This is one of the clearest findings in the project. The classical water fraction appears systematically higher, suggesting that

its threshold rules may classify reflections, shadows, blue-gray surfaces, or smooth bright regions as water. The DL method estimated lower water fractions overall, possibly because it separates reflected objects, shoreline, snow/ice, or ground from actual water more conservatively.

The year-level results suggest that the difference between the two methods is not random. Classical water estimates are consistently higher, while DL water estimates are lower and more compressed. This indicates a systematic difference in how the methods define and detect water.

5. Error Analysis

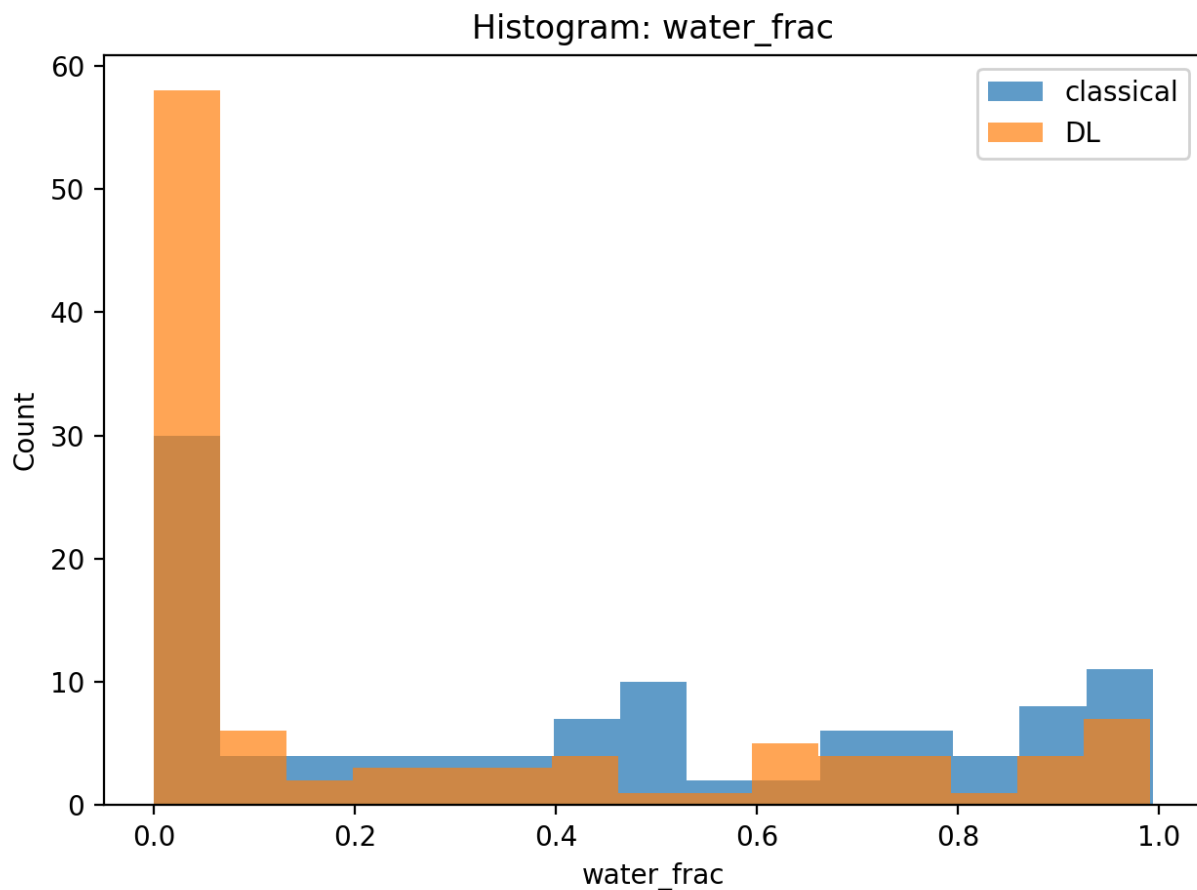


Figure 5. Histogram of water fraction. The classical method's distribution is shifted toward higher water fractions compared with DL, suggesting possible over-detection of water-like surfaces.

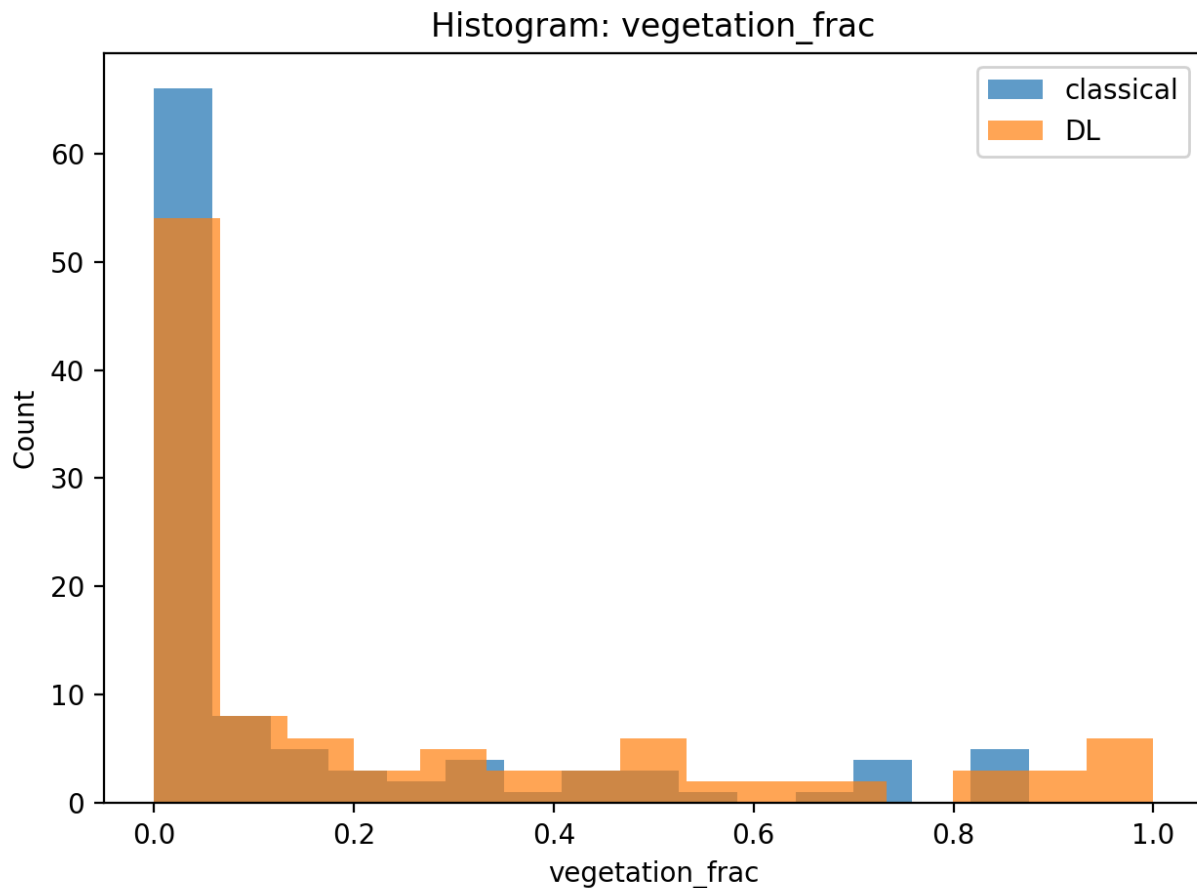


Figure 6. Histogram of vegetation fraction. The difference between distributions reflects the classical method’s difficulty with non-green winter vegetation and the DL model’s tendency to classify tree structure semantically.

The outlier audit examined images with the largest disagreements between the classical and DL methods. The goal was to understand why the models disagreed, not just report that they disagreed.

Several recurring failure modes appeared.

5.1 Reflections

Reflection was a major failure mode for water detection. In some images, calm water reflected trees, shorelines, or sky. The classical method sometimes under-counted water when the reflected water surface was muted and mixed with brown shoreline colors. In other cases, it may have over-counted water because reflective regions and dark mirrored shapes matched water-like thresholds. The DL model also struggled because reflections create semantic ambiguity: reflected trees can look like vegetation, but they are physically part of the water surface.

This means water detection is not simply a color problem. It requires deciding whether reflected objects should be counted as water or as the object being reflected. Classical thresholding and semantic segmentation answer that question differently.

5.2 Brown or Leafless Vegetation

Brown vegetation was the clearest failure mode for vegetation detection. Many eagle photographs contain winter trees, bare branches, gray trunks, or brown brush. The classical method often under-counted these areas because they do not satisfy green vegetation thresholds. The DL model often counted more vegetation because it recognized tree/brush structure semantically, even when the vegetation was not green.

This explains why DL vegetation estimates were often higher than classical vegetation estimates. The DL model appears better at identifying non-green vegetation structure, but it may also over-count bare winter branches as vegetation depending on the interpretation of “vegetation fraction.”

5.3 Sky-Dominated Scenes

Some eagle images were dominated by sky, especially when the eagle was flying or perched high in a tree. In these cases, both water and vegetation estimates can become **unstable** because the actual habitat area is small. Even with the bottom-65% ROI, some images still contained large amounts of sky. Classical thresholding may produce very low habitat fractions in these images, while DL may label small branches, cranes, or edge structures as semantic objects.

This suggests that an eagle photo is not always a good habitat photo. Some images are mainly subject/sky compositions, not environmental context images.

5.4 Snow and Ice

Snow and ice introduced additional ambiguity. Bright, low-saturation snow can interfere with classical thresholds, while frozen water creates a semantic problem for DL: ice may be part of a water body, but visually it can resemble ground, snow, or other bright surfaces. In one audited example, frozen/partially frozen water and snowy shoreline caused disagreement because the classical method did not consistently identify the bright gray surface as water, while the DL model treated much of it as water-like or semantically connected to the water body.

This shows that winter scenes require special handling. A future version of the project should include a separate snow/ice metric and possibly treat frozen water differently from liquid water.

5.5 Shoreline Ambiguity

Shoreline scenes were difficult because land, water, ice, mud, rocks, and reflections often appear in the same small area. Mixed pixels at boundaries can cause threshold leakage for the classical method, while DL segmentation may make different semantic decisions depending on whether a region looks like water, ground, or vegetation. Shoreline ambiguity is especially important in eagle habitat photographs because many eagle images are taken near rivers, lakes, or banks.

6. Discussion

The main finding is that classical thresholding and deep semantic segmentation produce systematically different environmental measurements on the same eagle photo dataset. The methods agree somewhat for vegetation, but even there the agreement is limited. For water, the disagreement is larger and more systematic.

This result matters because image-based environmental measurements are not neutral. The model used to define “water” or “vegetation” directly changes the numerical result. A simple classical method may be interpretable and fast, but it can be overly sensitive to color, brightness, reflections, and seasonal changes. A deep learning method can better identify semantic structures, but it can also introduce model-specific assumptions and may not perfectly handle unusual wildlife-photo scenes.

Therefore, the project should not be interpreted as proving a specific environmental trend. Instead, it demonstrates a reusable pipeline and a methodological result: different segmentation approaches can produce different ecological measurements, and those differences can be explained through failure-mode analysis.

The most defensible conclusion is that DL segmentation is generally more useful for semantic vegetation structure, especially in winter or leafless scenes, while classical thresholding is useful as a transparent baseline but is vulnerable to reflections, sky dominance, and color shifts. For water detection, neither method should be treated as ground truth without manual validation, because water reflections and shoreline ambiguity caused major disagreements.

7. Limitations

This project has several limitations.

First, the **dataset is heterogeneous**. The photographs vary in framing, distance, lighting, season, and scene composition. This makes it difficult to interpret year-level differences as true environmental change. For example, one year may contain more sky-heavy or water-heavy photos simply because of how the eagle was photographed.

Second, there is **no hand-labeled ground truth**. Without manual segmentation labels, the project can compare model agreement and disagreement, but it cannot definitively say which method is more accurate for every image.

Third, the DL model was not trained specifically for eagle habitat photographs. It may not handle wildlife-specific compositions, frozen water, reflections, or small shoreline features perfectly.

Fourth, the classical method is deliberately simple. Its value is interpretability, but it lacks the semantic understanding needed for complex scenes.

8. Future Work

The most important next step is to create a small manually labeled validation set. For example, 20–50 images could be manually labeled for water and vegetation using rough masks or category bins. That would allow accuracy evaluation rather than only method comparison.

Future work should also group images by scene type, such as sky-dominated, water-dominated, snowy/icy, forested, and shoreline scenes. Agreement between classical and DL methods could then be measured within each category. This would show which environments are easiest or hardest for each method.

Another improvement would be to exclude the eagle itself from the analysis using object detection or manual masking. Since the project aims to analyze environmental context, the subject can sometimes interfere with background composition metrics.

Finally, the dataset could be expanded with more consistent photos from the same site, season, and angle. That would make environmental trend analysis more defensible.

9. Conclusion

This project built and evaluated a computer vision pipeline for analyzing environmental context in eagle habitat photographs. Across 106 images, classical color-threshold segmentation and deep semantic segmentation produced different estimates of vegetation and water coverage. Vegetation estimates showed moderate but limited agreement, while water estimates showed weak agreement and large systematic differences. The error analysis showed that disagreements were not random; they were tied to

recognizable visual conditions such as reflections, brown winter vegetation, sky-heavy images, snow/ice, and shoreline ambiguity.

The final result is a stronger project than a simple trend chart because it evaluates measurement reliability. Rather than claiming that one model automatically reveals environmental change, the project shows how different computer vision methods define and detect environmental characteristics differently. This makes the project useful as a prototype for future habitat-image analysis and as a demonstration of careful, honest applied machine learning methodology.

Data and Code Availability

The final project folder contains the generated analysis outputs, including the comparison summary CSV, year-level means CSV, merged per-image comparison table, disagreement audit table, qualitative audit notes, and final figures. The project scripts, image-processing outputs, and report files were stored locally in the project directory. The final organized package is located in the results_final folder, which includes the PDF report, supporting CSV files, audit files, and figure images. The project code and selected outputs will also be uploaded to a GitHub repository for documentation and reproducibility.

References

- Hugging Face. (2026). SegFormer. Hugging Face Transformers documentation. https://huggingface.co/docs/transformers/en/model_doc/segformer
- Pandas Development Team. (2026). pandas: Python data analysis library. <https://pandas.pydata.org/>
- Python Software Foundation. (2026). Python language reference. <https://www.python.org/>
- PyTorch Foundation. (2026). PyTorch documentation. <https://docs.pytorch.org/docs/stable/index.html>
- Xie, E., Wang, W., Yu, Z., Anandkumar, A., Alvarez, J. M., & Luo, P. (2021). SegFormer: Simple and efficient design for semantic segmentation with transformers. *Advances in Neural Information Processing Systems*, 34. <https://arxiv.org/abs/2105.15203>
- Zhou, B., Zhao, H., Puig, X., Fidler, S., Barriuso, A., & Torralba, A. (2017). Scene parsing through ADE20K dataset. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. https://openaccess.thecvf.com/content_cvpr_2017/html/Zhou_Scene_Parsing_Through_CVPR_2017_paper.html

Appendix A: Short Portfolio Description

I built a computer vision pipeline to analyze environmental context in 106 eagle habitat photographs. The project compares a classical color-threshold segmentation method with a deep learning semantic segmentation model to estimate water and vegetation coverage in each image. Instead of treating one model as automatically correct, I evaluated how the two methods disagreed and used visual error analysis to identify failure modes such as reflections, winter vegetation, sky-heavy compositions, snow/ice, and shoreline ambiguity. The project demonstrates applied machine learning, image processing, data validation, and critical interpretation of model outputs.